

Gas Cylinder Filling Safety Control System

Engineering Design, Safety Architecture & Digital Prototype Plan

Consolidated Project Document — Revision 3
Merges the Design Baseline and the Project Status Report into one record

Consolidated engineering record for a safety-oriented control system for O₂/N₂ cylinder filling. Covers process understanding, hazard analysis, safety requirements, safety and electrical architecture, component requirements, and the digital prototype and test plan that precedes any physical high-pressure integration.

ENGINEERING NOTE

120 bar and 125 bar are Company-Approved Setpoints for this project. They are not independently validated legal or engineering limits and must be checked against actual equipment specifications, applicable standards, and qualified technical/safety approval before installation.

DOCUMENT CONTROL

Field	Value
Revision	3
Consolidates	Design Baseline (Rev. 2) and Project Status Report into a single document
Current stage	Digital Prototype & Simulation preparation (Stage 9 of 21)
Design principle	Validate the complete safety logic digitally before physical prototyping or real-equipment integration
Status	Draft — pending detailed component selection and qualified safety/engineering approval

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1. Executive Summary

The project is a safety-oriented control system for a gas-cylinder filling process. The design separates the safety function from monitoring and operator-interface functions and defines a fail-safe operating philosophy. The system detects excessive pressure, sensor faults, valve faults, emergency-stop activation, power loss, and controller failures, and prevents uncontrolled filling.

Protection is layered: a passive mechanical relief valve and the existing ramp gauge (PG1) act independently of the electronic chain, which itself is backed by a physical emergency stop. Section 8 sets out this defense-in-depth architecture.

Process understanding, hazard analysis, safety requirements, system architecture, hardware architecture, electrical architecture, and the conceptual circuit architecture are complete. Component technical requirements and I/O and terminal architecture are defined. The next major activity is construction of a digital prototype, followed by systematic functional, safety, and fault-injection testing, before any physical connection to a high-pressure gas system.

2. Project Development Roadmap

Stage	Activity	Status
1	Process Understanding	Completed
2	Hazard Analysis	Completed
3	Safety Requirements Specification (SRS)	Completed
4	System Architecture	Completed
5	Hardware Architecture	Completed — relief valve added
6	Circuit & Wiring Architecture	Concept completed
7	Detailed Safety Circuit Design	Defined
8	Detailed Hardware & Component Selection	Requirements defined
9	Digital Prototype / Simulation	Next
10	Digital Functional & Safety Testing	Next
11	Firmware Development	Planned
12	Monitoring / Logging Software	Planned
13	Integrated Digital Prototype	Planned
14	Full Digital Validation & Fault Injection	Planned
15	Physical Prototype	Planned
16	Physical Bench Testing	Planned
17	Real Equipment Integration	Planned
18	Commissioning & Site Testing	Planned

Stage	Activity	Status
19	Final Safety Validation	Planned
20	Production Deployment	Planned
21	Final Documentation	Planned

3. Real Process and Equipment

- The shop fills Oxygen (O₂) and Nitrogen (N₂).
- There are 8 large source cylinders, stated for this project as 60 L and 160 bar.
- Each source cylinder has its own hose/path and a separate manually operated valve V1–V8.
- Normally, two source-cylinder valves are never open simultaneously.
- The Ramp/Manifold is factory-made, standardized and government-approved; this project does not redesign the Ramp.
- The Ramp carries the existing analog gauge PG1 and, per Section 8, an added mechanical relief valve.
- The filling hose terminates in a filling clamp for the customer cylinder.
- Filling proceeds sequentially from the lower-pressure source cylinder toward the higher-pressure source cylinder; source cylinder #8 is stated to be above 120 bar and can complete the fill.
- The operator opens one source-cylinder valve, waits for pressure transfer/stabilization, closes it, then moves to the next.

4. Cylinder Approval Boundary

- Before filling, the operator checks the test sheet and physical appearance.
- The operator remains solely responsible for deciding whether a cylinder is eligible for filling.
- The system does not inspect test dates, certification, corrosion, dents, physical condition or other approval criteria.
- The safety-system scope begins after the operator has approved the cylinder and filling has started.

5. Company-Approved Setpoints

Setpoint	Meaning	System Action
120 bar	Warning / Target Reached	Warning/target indication; not an emergency shutdown threshold.
125 bar	Emergency Shutdown	Initiate shutdown, close Smart Main Valve, disable filling and enter LOCKED/FAULT.

The 120→125 bar span is adequate because the approach to final pressure is slow: filling is manual and sequential through up to 8 source cylinders, with the operator waiting for pressure stabilization between valves. This basis should be re-confirmed once actual valve closure time and worst-case flow rate are known.

6. Physical Process Architecture

Element	Status	Notes
C1–C8	8 Source Cylinders	60 L / 160 bar each, individually valved V1–V8
Ramp / Manifold	Existing, unmodified	Carries PG1 and the added mechanical relief valve
Smart Main Valve Assembly (SMVA)	New	Electric shutoff valve, pressure transducer, digital gauge, at the existing Main Valve location

Element	Status	Notes
Filling Hose → Filling Clamp	Existing, unmodified	Terminates at the customer cylinder

Flow path: C1–C8 (via V1–V8) → Common Ramp/Manifold (PG1, relief valve) → Smart Main Valve Assembly → Filling Hose → Filling Clamp → Customer Cylinder.

7. Smart Main Valve Assembly (SMVA)

The existing Main Valve location is replaced by a Smart Main Valve Assembly. The approved Ramp itself remains unchanged.

Component	Function
Electrically controlled high-pressure shutoff valve	Primary automated actuator; closes on emergency shutdown or valve fault
Pressure transducer (PT-01)	4–20 mA electronic measurement for controller/safety logic
Digital pressure gauge	Operator-facing indication; not the sole safety decision source

- Automatic emergency shutdown capability.
- The company will provide suitable equipment for O₂/N₂ and high-pressure service.
- **Oxygen service scope:** wetted-material compatibility, cleaning/degreasing standard, and ignition-risk considerations for O₂ service are addressed during detailed equipment selection, not in this document.

8. Defense-in-Depth Safety Architecture

A single electronic measurement-and-shutdown chain is not sufficient for a hazard rated Critical. Four layers are in place so that no single point of failure, mechanical, electronic, or human, can cause uncontrolled continuation of filling.

#	Element	Type	Trigger	Independence Rationale
1	Mechanical relief valve	Passive / mechanical	Automatic	Independent of controller, power, and software. Vents on overpressure regardless of electronic system state.
2	Ramp gauge PG1	Mechanical / visual	Operator-observed	Non-electronic pressure reference with no shared failure mode with the transducer/controller/display chain. Lets the operator visually cross-check the digital display, especially during Warning state.
3	Pressure transducer + Safety Controller	Electronic	Automatic	Primary automated safety measurement and shutdown logic: Pressure Transducer → Safety Controller → Safety Logic → Valve Control → Shutoff Valve.
4	Physical E-Stop	Electromechanical	Manual, operator-initiated	Independent of normal software execution: E-Stop → Safety Shutdown Path → Valve Safe State.

Layer 3 is the only layer that depends on power, firmware, and a single sensor. Layers 1 and 2 share no failure mode with it: the relief valve is purely mechanical, and PG1 is a direct analog reading with its own sensing path. A silent transducer drift or controller fault removes the automated layer only; the mechanical and visual layers stay intact.

DEFINED SAFE STATE

Filling disabled, shutoff valve in its safe position, system locked, no automatic restart. Applies to emergency conditions, E-stop activation, sensor faults, valve faults, controller faults, power failure, and safety-circuit faults.

9. Hazard Analysis

Hazard	Initial Severity
Overpressure	Critical
Pressure gauge failure	Critical
Pressure sensor failure	Critical
Shutdown valve/device failure	Critical
Two paths accidentally open simultaneously	High
Wrong source-cylinder selection	High
Power failure	High
Sensor/cable failure	High
Abnormal pressure rise	High
Operator error	Critical
Controller/watchdog failure	Critical

CORE SAFETY PRINCIPLE

Human operation remains present, but an automated safety layer monitors pressure and can force the filling path to a safe state. No single point failure should cause uncontrolled continuation of filling.

10. Safety Requirements Specification (SRS)

- **SR-01** — Continuously measure/monitor pressure.
- **SR-02** — Display current pressure to the operator.
- **SR-03** — Detect pressure increase/decrease.
- **SR-04** — At 120 bar activate Warning/Target Reached indication.
- **SR-05** — At ≥ 125 bar initiate Emergency Shutdown.
- **SR-06** — After Emergency Shutdown enter LOCKED/FAULT.
- **SR-07** — No automatic restart after pressure drops or controller restarts.
- **SR-08** — Reset requires intentional authorized action via a physical key switch; Reset does not automatically start filling.
- **SR-09** — Detect sensor disconnect, invalid/out-of-range signal, frozen/unresponsive signal and electrical fault.
- **SR-10** — Sensor failure must not permit uncontrolled filling; transition to safe state and disable filling.
- **SR-11** — Power loss must not cause unsafe continuation; exact hardware behavior is defined during detailed design.
- **SR-12** — Startup: Power ON → Self Check → READY → operator action → START. No auto filling.
- **SR-13** — Physical Emergency Stop causes safety action, disables filling and locks the system.

- **SR-14** — Log Start, Stop, Warning, Emergency Shutdown, Sensor Fault, Power Failure, Reset and Maximum Pressure.
- **SR-15** — Mechanical relief valve on the Ramp provides passive overpressure protection independent of controller state, power availability, and software logic.
- **SR-16** — Start is only a request: the controller verifies E-stop status, sensor health, valve status, system readiness, and absence of faults before filling can begin.

GOLDEN PRINCIPLE

No single point failure should cause uncontrolled continuation.

11. System State Machine

State	Description / Transition
POWER ON	Initial system energization → SELF CHECK
SELF CHECK	Verification of safety conditions. OK → READY. FAULT → LOCK
READY	System healthy, waiting for start request. START → FILLING
FILLING	Controlled filling operation. $P < 120$ → continues
WARNING	$120 \leq P < 125$ bar, target-reached indication
EMERGENCY	$P \geq 125$ bar → VALVE CLOSED → LOCK
SENSOR FAULT	Pressure measurement invalid/unavailable → SAFE STATE → LOCK
VALVE FAULT	Command/feedback inconsistent → SAFE RESPONSE → LOCK
E-STOP	Emergency stop activated → SHUTDOWN → LOCK
POWER FAILURE	Loss of control power → SAFE STATE → LOCK
LOCK	Latched safe condition. Authorized RESET (key switch) → SELF CHECK → READY

No direct LOCK → FILLING transition. Reset only returns to READY; Start is required separately.

12. Safety and Monitoring Architecture

Safety chain

Pressure Transducer → Safety Controller → Safety Logic → Valve Control → Shutoff Valve.

Independent E-stop path

E-Stop → Safety Shutdown Path → Valve Safe State. Independent of normal software execution.

Monitoring layer

Controller → Display / Alarm / Logging / UI.

- Ramp gauge PG1 remains in place and unmodified, providing an independent, non-electronic pressure reference the operator can visually cross-check against the digital display.
- Safety layer handles dangerous-condition detection, emergency shutdown, fault handling and lockout.
- Monitoring layer handles pressure display, status, warning, logging and reporting.
- Monitoring software must not override safety shutdown.
- PC, internet or Python must not be the sole mechanism responsible for emergency shutdown.

13. Hardware Architecture

13.1 Pressure Transducer (PT-01)

- Primary electronic pressure measurement, 4–20 mA.
- Selection parameters: pressure range, accuracy (preferably $\leq \pm 0.5\%$ FS), pressure rating, O₂/N₂ compatibility, wetted materials, oxygen-service preparation and diagnostics.
- Generic conversion: $P = P_{min} + ((I - I_{min}) / (I_{max} - I_{min})) \times (P_{max} - P_{min})$.

13.2 Digital Pressure Gauge

- Operator-facing display of pressure and status; not the sole safety decision source, cross-checked against the retained analog gauge PG1.

13.3 Safety Controller

- Industrial/safety-capable controller with required AI/DI/DO, diagnostics, watchdog, safety-rated architecture, valve control and E-stop integration.

13.4 Electric Shutoff Valve

- Replaces the existing Main Valve function at the same location.
- High-pressure rated, suitable for O₂/N₂ service as applicable; defined actuation and fail-safe behavior; position feedback preferred.

13.5 Valve Position Feedback

CLOSE command → Electric Valve → Position Feedback → Controller.

Command ≠ feedback → VALVE FAULT → Safety Response → LOCK. Both OPEN and CLOSED active simultaneously is also a fault unless the valve manufacturer defines another valid interpretation.

13.6 Mechanical Relief Valve

- Passive mechanical relief/safety valve mounted on the Ramp, independent of the electronic control path.
- Vents automatically on overpressure with no dependency on controller power, firmware, or logic state.
- Proposed during design review and accepted by the company; sizing and set pressure are pending equipment selection and qualified safety/engineering approval.

13.7 Power and Watchdog

- Control power: controller, sensor, display and indicators. Actuator power: electric shutoff valve.
- Power-failure behavior must prevent unsafe continuation and automatic restart.
- Watchdog detects controller/firmware failure and initiates the defined safe response.

14. Electrical Architecture

The electrical design is based on a protected AC input feeding a control power supply and a separately considered valve power path. Exact supply rating and protection values remain dependent on final component selections.

14.1 Power Distribution

AC Input → Main Isolator → Main Protection → Control Power Supply → 24 VDC Control Circuit.

The 24 VDC control circuit supplies the controller, pressure transducer, display, operator controls, indicators, and associated control devices. Valve power is kept separately identifiable to prevent actuator faults from compromising the control supply.

14.2 Grounding and Protection

- Protective Earth (PE) is provided for the control enclosure, metallic equipment, and applicable exposed conductive parts.
- Signal 0V and protective earth are treated as separate design considerations, finalized with the selected equipment and panel design.
- AC and DC protection will be sized after actual power consumption, inrush current, cable ratings, and device requirements are known.

15. Preliminary I/O List

Tag	Type	Function
AI-01	Analog	PT-01 pressure transducer, 4–20 mA — main pressure measurement

Tag	Type	Function
DI-01	Digital	E-Stop status
DI-02	Digital	Start request
DI-03	Digital	Reset request (key switch)
DI-04	Digital	Valve OPEN feedback
DI-05	Digital	Valve CLOSED feedback
DI-06	Digital	External fault

Tag	Type	Function
DO-01	Digital	Shutoff valve command
DO-02	Digital	Warning lamp
DO-03	Digital	Buzzer
DO-04	Digital	Emergency lamp
DO-05	Digital	Ready lamp
DO-06	Digital	Filling lamp

The mechanical relief valve and PG1 are passive/mechanical and are not wired into controller I/O; they operate independently of the electronic system by design.

16. Wiring and Terminal Architecture

Field devices are routed through terminal blocks before reaching the controller, providing a clear interface for commissioning, testing, troubleshooting, maintenance, and future replacement or expansion.

16.1 Pressure Transducer

PT-01 is a 4–20 mA pressure transmitter. The two-wire or three-wire connection and input return arrangement will be finalized from the selected controller and transmitter datasheets.

16.2 Valve Command and Feedback

The controller output is routed through a valve driver/interface to the shutoff valve. Open and closed position feedback are returned to the controller when supported by the final valve assembly.

A command/feedback mismatch is a valve fault. Both OPEN and CLOSED active simultaneously is also a fault unless the valve manufacturer defines another valid interpretation.

17. Pressure Decision Logic

Condition	System Response
$P < 120 \text{ bar}$	Normal filling operation
$120 \leq P < 125 \text{ bar}$	Warning / target reached indication
$P \geq 125 \text{ bar}$	Emergency shutdown, valve close, filling disabled, system lock
Invalid / disconnected / out-of-range sensor	Sensor fault, safe state, system lock

18. Sensor Fault Logic

- Disconnected sensor.
- Invalid electrical signal.
- Out-of-range measurement.
- Frozen/stuck/unresponsive signal.
- Electrical fault.
- Where feasible, implausible pressure behavior.

Sensor Fault → Safe Response → Filling Disabled → LOCK.

PG1 remains available to the operator as a visual reference during a transducer/controller sensor fault.

19. Valve Fault Logic

If feedback is available: Command $\neq$ Actual Position $\rightarrow$ VALVE FAULT $\rightarrow$ Safety Response $\rightarrow$ LOCK.

20. Power Failure Behavior

During filling: FILLING $\rightarrow$ POWER FAILURE $\rightarrow$ SAFE STATE $\rightarrow$ LOCK / FILLING DISABLED.

After restoration: POWER RESTORED $\rightarrow$ SELF CHECK $\rightarrow$ READY.

No automatic restart. The mechanical relief valve remains active during a power failure, since it does not depend on electrical power.

21. Component Requirements

Component	Key Requirements	Status
Pressure Transducer	4–20 mA; suitable pressure range; accuracy preferably $\leq \pm 0.5\%$ FS; compatible materials and gas service; appropriate overpressure rating	Requirements defined
Digital Pressure Gauge	Clear local indication; suitable range and resolution; gas-service compatible; independent of safety decision	Requirements defined
Safety Controller	Industrial controller with required AI/DI/DO; diagnostics; watchdog; safety-rated architecture where required	Selection pending
Electric Shutoff Valve	High-pressure rated; suitable for O ₂ /N ₂ service as applicable; defined actuation and fail-safe behavior; position feedback preferred	Selection pending
Mechanical Relief Valve	Passive overpressure relief, independent of controller/power; set pressure coordinated with the 125 bar electronic setpoint	Selection pending
E-Stop	Physical, mechanically latching emergency-stop device with suitable contacts and safety-circuit compatibility	Selection pending
Power Supply	Industrial 24 VDC control supply; capacity and protection calculated from final loads	Capacity pending
Protection	Appropriate AC/DC overcurrent and circuit protection	Sizing pending
Enclosure / Terminals	Industrial panel and terminal system suitable for the environment and wiring architecture	Selection pending

22. Event Logging

- System Start
- System Stop
- Filling Start
- Filling Stop
- 120-bar Warning
- Emergency Shutdown
- Sensor Fault
- Valve Fault
- Emergency Stop
- Power Failure
- Reset (with operator/key identifier if available)
- Maximum Pressure

Potential future fields: date/time, filling cycle ID, duration, fault cause, system state and operator ID if required.

23. Preliminary Bill of Materials (BOM)

No.	Component	Qty.
1	High-Pressure Electric Shutoff Valve	1
2	Pressure Transducer (PT-01)	1
3	Digital Pressure Gauge / Display	1
4	Safety Controller	1
5	Emergency Stop	1 or more
6	Warning Lamp	1
7	Buzzer	1
8	Reset Key Switch	1
9	Start Button	1
10	Status Indicators	1 set
11	Power Supply	1
12	Electrical Protection Components	1 set
13	Valve Position Feedback	1
14	Mechanical Relief / Safety Valve	1
15	Control Panel / Enclosure	1
16	Terminal / Wiring Components	1 set

24. Digital Prototype Strategy

No physical high-pressure connection is planned at this stage. The next development activity is a digital prototype that models the pressure source, pressure transducer, controller, safety logic, valve behavior, filling process, alarms, and operator interface.

24.1 Digital System Model

Virtual Pressure Source → Virtual Pressure Sensor → Controller Model → Safety Logic → Virtual Shutoff Valve → Filling Process Model.

The digital prototype also provides a monitoring interface showing pressure, system state, valve state, sensor status, E-stop status, alarms, and event history.

The mechanical relief valve and PG1 are passive/mechanical elements validated by physical inspection and calibration, not by digital simulation; they are out of scope for the digital test matrix in Section 25.

25. Digital Safety Test Plan

Test ID	Condition	Expected Response
T-01	Pressure = 119 bar	Normal operation
T-02	Pressure = 120 bar	Warning
T-03	Pressure = 124.9 bar	Warning
T-04	Pressure $\geq$ 125 bar	Emergency shutdown + valve safe state + lock
T-05	Pressure spike	Safety logic responds without uncontrolled continuation
T-06	Sensor disconnected	Sensor fault + safe state + lock
T-07	Sensor out of range	Sensor fault + safe state + lock
T-08	Sensor frozen	Fault detection per defined diagnostic criteria
T-09	Valve fails to close	Valve fault + filling disabled + lock
T-10	Valve feedback mismatch	Valve fault + lock
T-11	E-Stop during filling	Immediate safety shutdown path + lock
T-12	Power failure during filling	Safe state; filling disabled
T-13	Power restored	Self-check $\rightarrow$ READY; no automatic restart
T-14	Controller/watchdog failure	Safety response through the selected architecture
T-15	Reset after fault	Self-check required before READY
T-16	Start from READY	Filling begins only when all preconditions are satisfied
T-17	Invalid startup condition	Start rejected; no filling

26. Planned Digital Prototype Features

- Pressure simulation from zero through the normal, warning, and emergency regions.
- Virtual sensor behavior including valid signal, disconnect, out-of-range, and frozen-signal conditions.
- Safety state machine with explicit transitions and latched fault handling.
- Virtual valve model with command, transition, open/closed feedback, and failure modes.
- Emergency-stop simulation independent of the normal monitoring interface.
- Power-loss and recovery simulation with no automatic restart.
- Watchdog/controller-failure simulation.
- Alarm, status indication, and event logging.
- Automated execution of the digital test matrix.
- Fault injection to demonstrate safe responses before any physical prototype is built.

27. Engineering Decisions and Open Items

The architecture is intentionally not locked to a specific manufacturer or model at this stage. Final component selection depends on verified requirements for the Smart Main Valve Assembly, pressure-transducer range and connection, actuator voltage, valve fail-safe behavior, environmental conditions, applicable safety requirements, and mechanical relief valve set pressure.

The fail-safe position and actuation method of the main shutoff valve must be established from the actual valve design and its certified/approved application. The safety function must not depend solely on application software.

28. Current Project Status

Area	Status
Process and hazard definition	Completed
Safety requirements and safe-state philosophy	Completed
System and hardware architecture	Completed — relief valve added
Safety circuit architecture	Completed at design level
Electrical architecture	Completed at design level
I/O and terminal architecture	Defined
Component technical requirements	Defined
Final component models	Pending verified selection
Physical wiring	Not started
High-pressure integration	Not started
Digital prototype	Next major activity

29. Next Planned Step

The immediate next step is Phase 6.6, Digital Prototype & Simulation. The prototype will be developed in Python as a controlled engineering model, with a pressure simulator, sensor model, safety state machine, valve model, fault injection, event logging, and a basic operator interface. The complete digital test matrix (Section 25) will be executed before any physical high-pressure integration is considered.

Development sequence: Digital Model → Digital Safety Testing → Digital Validation → Physical Prototype → Bench Testing → Technical Approval → Real Equipment Integration → Commissioning → Final Safety Validation → Production Operation.

30. Non-Negotiable Project Principles

- Do not redesign the factory-approved Ramp.
- Replace the existing Main Valve function with the Smart Main Valve Assembly at the Main Valve location.
- Use company-provided, appropriately rated high-pressure/O₂/N₂ equipment.
- Operator remains solely responsible for cylinder eligibility/approval.
- Safety-system scope starts once filling begins.
- Monitoring software cannot override safety shutdown.
- No automatic restart after shutdown, fault or power restoration.
- Reset ≠ Start; reset is performed via a physical key switch.
- 120 bar = Company-Approved Warning / Target. 125 bar = Company-Approved Emergency Shutdown.
- Safety-critical logic should be independent from PC/internet/UI as practical.
- Valve position feedback is strongly preferred.
- Pressure transducer is the safety measurement source; digital gauge is primarily operator indication.
- Existing Ramp gauge PG1 remains in place and unmodified, serving as an independent, non-electronic pressure reference for the operator.
- A mechanical relief valve is added to the Ramp as a passive overpressure protection layer, independent of the electronic control path; an addition, not a modification, to the existing approved Ramp.
- No physical high-pressure connection until the digital prototype passes the complete test matrix.
- Final installation requires validation against actual equipment, applicable standards and qualified safety/engineering approval.

End of Consolidated Project Document, Revision 3. Subject to detailed component selection and qualified safety/engineering approval.